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Studierichting: Informatica
Oefeningenreeks 5E nr. 1

$$\begin{cases} x(t) = t^2 \\ y(t) = 4t - t^3 \end{cases}$$

$$S = \int_{t_1}^{t_2} |4t - t^3|(t^2)'dt$$

$$= \int_{t_1}^{t_2} |4t - t^3|2tdt$$

$$\int |4t - t^3|2tdt$$

$$= \int (8t^2 - 2t^4)dt$$

$$= 8\int t^2dt - 2\int t^4dt$$

$$= \frac{8t^3}{3} - \frac{2t^5}{5} + c$$

$$|4t - t^3| = 0 \vee t^2 = 0$$

$$\Leftrightarrow |t(4 - t^2)| = 0 \vee t = 0$$

$$\Leftrightarrow t = -2 \vee t = 0 \vee t = 2$$

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$$2\int_0^2 |4t - t^3|2tdt$$

$$= 2\left[\frac{8t^3}{3} - \frac{2t^5}{5}\right]_0^2 = 2\cdot\left(\frac{8\cdot 2^3}{3} - \frac{2\cdot 2^5}{5}\right) - 2\cdot 0 = \frac{640}{15} - \frac{384}{15} = \frac{256}{15}$$

References